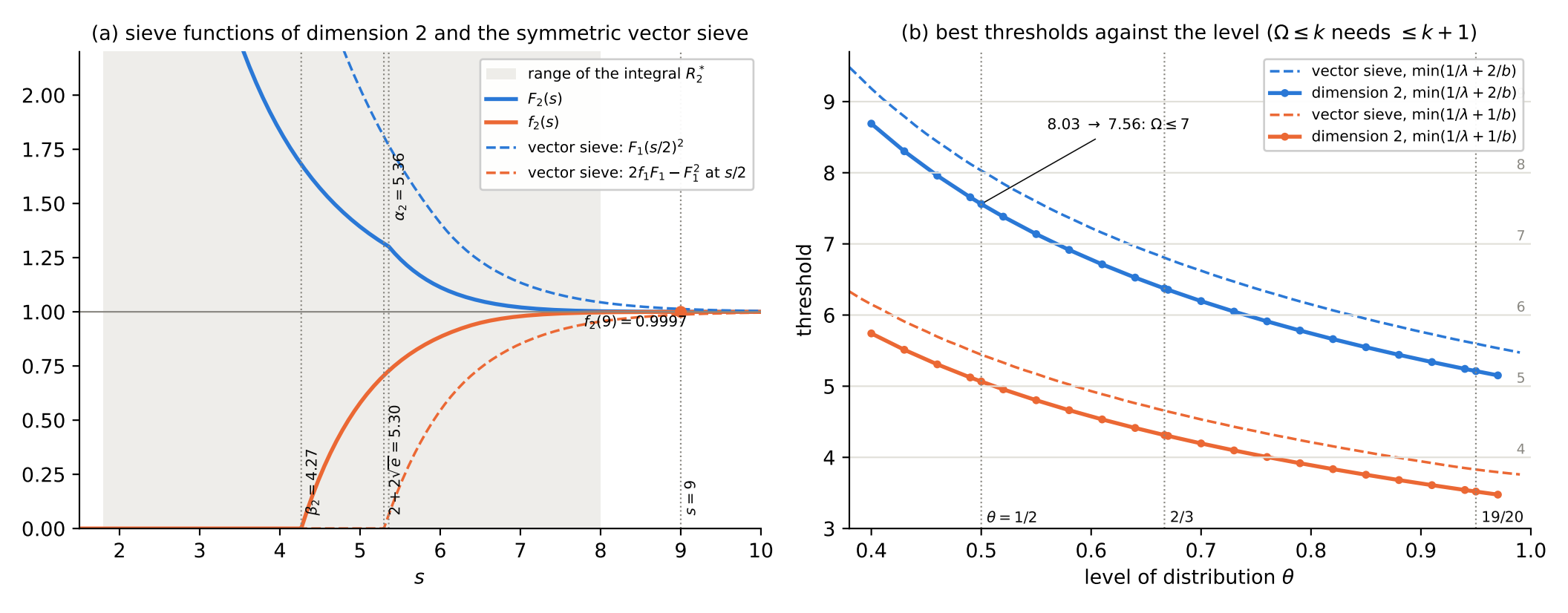
(a) sieve functions of dimension 2 and the symmetric vector sieve
range of the integral R_2^*
$F_2(s)$
$f_2(s)$
vector sieve: $F_1(s/2)^2$
vector sieve: $2f_1F_1 - F_1^2$ at $s/2$
$\alpha_2 = 5.36$
$\beta_2 = 4.27$
$2 + 2\sqrt{e} = 5.30$
$s = 9$
$f_2(9) = 0.9997$
s
(b) best thresholds against the level ($\Omega \le k$ needs $\le k+1$)
vector sieve, $\min(1/\lambda + 2/b)$
dimension 2, $\min(1/\lambda + 2/b)$
vector sieve, $\min(1/\lambda + 1/b)$
dimension 2, $\min(1/\lambda + 1/b)$
8.03 → 7.56: $\Omega \le 7$
threshold
level of distribution θ
$\theta = 1/2$
2/3
19/20